By TOUCH DEN · November 11, 2026

Common Attacks



Video 1 · How Attacks Start, and Attacking People

Security Fundamentals · Section 5

In this video, you will learn:

- The three ways most attacks begin
- Phishing and social engineering
- How to recognize and resist a phish

These attacks are explained to defend against. Only ever test systems you own or have written permission to test.

How most attacks start

Not a dramatic break-in. Small and quiet, almost always one of three doors:

1. ***A person is tricked*** *into clicking, opening, or telling.*
2. ***A weakness is exploited*** *in exposed, unpatched software.*
3. ***A stolen or guessed password*** *is used to simply log in.*

The attacker needs one click, one unpatched server, or one reused password. ***Good news for defenders:*** *a few basic defenses on these three doors stop most attacks.*

Phishing and social engineering

Social engineering *attacks the person, not the computer.* ***Phishing*** *is the most common form: a fake message pretending to be someone you trust.*

- ***Spear phishing:*** *targeted and personalized.* ***Whaling:*** *aimed at a CEO.*
- ***Smishing / vishing:*** *by text / by call.*
- ***Business email compromise:*** *posing as a boss or supplier to redirect a payment, no malware at all.*

Signs: *urgency, a password or payment request, a slightly wrong sender, a link that lies about where it goes.*

Defending against phishing

- **Train** staff and keep a healthy suspicion.
- **Verify** the real sender address, and any urgent request, through another channel.
- **Never act on urgency** without checking.
- **Email filtering** to catch the obvious ones.
- **Multi-factor authentication**, so a stolen password alone is not enough.

Modern phishing can be flawless, so don't rely on spotting typos. Verify.

Recap: how attacks start

- Most attacks begin with a tricked person, an unpatched system, or a stolen password.
- Phishing attacks the person; it comes by email, text, and call.
- Verify the sender and the request; use MFA. Don't trust urgency.

Next video: malware and password attacks.

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Video 2 · Malware and Password Attacks

Security Fundamentals · Section 5

In this video, you will learn:

- The main kinds of malware
- How passwords are attacked
- The defenses for each

Malware: the main kinds

Malware is any software written to do harm. Behavior matters more than the label.

Type	How it behaves
Virus	Attaches to a file; spreads when a person runs/shares it
Worm	Spreads by itself across a network, no human action
Trojan	Pretends to be useful so you install it yourself
Ransomware	Locks/encrypts files and demands payment
Spyware	Secretly watches keystrokes, screens, passwords
Botnet	Turns your machine into one the attacker controls

One real sample often combines several.

Defending against malware

- ***Update software:*** *worms spread through known, unpatched holes.*
- ***Don't run untrusted programs or attachments.***
- ***Anti-malware tools.***
- ***Least privilege:*** *so malware can't reach everything.*
- ***Offline backups:*** *so ransomware can't destroy your only copy.*

Behavior, not the label, decides the defense.

Password attacks

- ***Brute force:*** *try millions of passwords.*
- ***Dictionary:*** *try common words and passwords first.*
- ***Credential stuffing:*** *reuse pairs leaked from one site on others, very effective, because people reuse passwords.*
- ***Spraying:*** *one common password against many accounts.*

*Attackers usually steal a **stored** password file and crack it **offline**, so how passwords are stored matters: slow **password hashing**, never plain text.*

Defenses: *unique passwords (a manager helps), **MFA**, strong hashing on the server.*

Recap: malware and passwords

- *Malware spreads differently (virus, worm, trojan); defend by behavior: updates, least privilege, offline backups.*
- *Passwords are mostly cracked offline; credential stuffing feeds on reuse.*
- *Unique passwords plus MFA defeat most password attacks.*

Next video: *attacks on services and networks, and the attack lifecycle.*

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Video 3 · Services, Networks, and the Attack Lifecycle

Security Fundamentals · Section 5

In this video, you will learn:

- Web and network service attacks
- Denial of service and man-in-the-middle
- The attack lifecycle, and why defenders have the edge

Web and network service attacks

Any internet-facing service is a door. Common web attacks:

- ***SQL injection:*** *input the server mistakes for a command, exposing the database.*
- ***Cross-site scripting (XSS):*** *malicious code that runs in other users' browsers.*
- ***Broken access control:*** *reaching data you shouldn't, e.g. another user's account.*

Common thread: *trusting input that shouldn't be trusted.* ***Defense:*** *treat all input as untrusted; the* ***server*** *enforces the rules. (See the OWASP Top 10.)*

Denial of service, and man-in-the-middle

Denial of service (DoS): *attacks* ***availability*** *by flooding a system so real users can't use it.* ***DDoS*** *sends the flood from many machines (a botnet), bigger and harder to block. Defense: capacity, filtering, anti-DDoS services.*

Man-in-the-middle (MITM): *the attacker secretly sits between you and the other side (classic on open Wi-Fi), reading or changing traffic. Defense: encryption, HTTPS, a VPN; heed certificate warnings.*

The attack lifecycle

Serious attacks move through stages (Cyber Kill Chain, MITRE ATT&CK):

Stage	What happens
Reconnaissance	Research the target
Initial access	Phish, exploit, or stolen password
Foothold	Keep access
Escalate privileges	Normal user → admin
Move laterally	Spread to valuable machines
Act on objective	Steal, encrypt, defraud

The attacker must win every stage. The defender only has to break the chain once.

Recap: services, networks, lifecycle

- *Service attacks come from trusting input; the server must enforce rules.*
- *DoS attacks availability; MITM reads unencrypted traffic, encryption defeats it.*
- *Attacks move through stages, and **defense in depth** breaks the chain at any one of them.*

Next lesson: *passwords and authentication, in depth.*